



MASTER THESIS: ANOMALY DETECTION METHODS FOR THE PREDICTION OF CONTRACTS TERMINATION

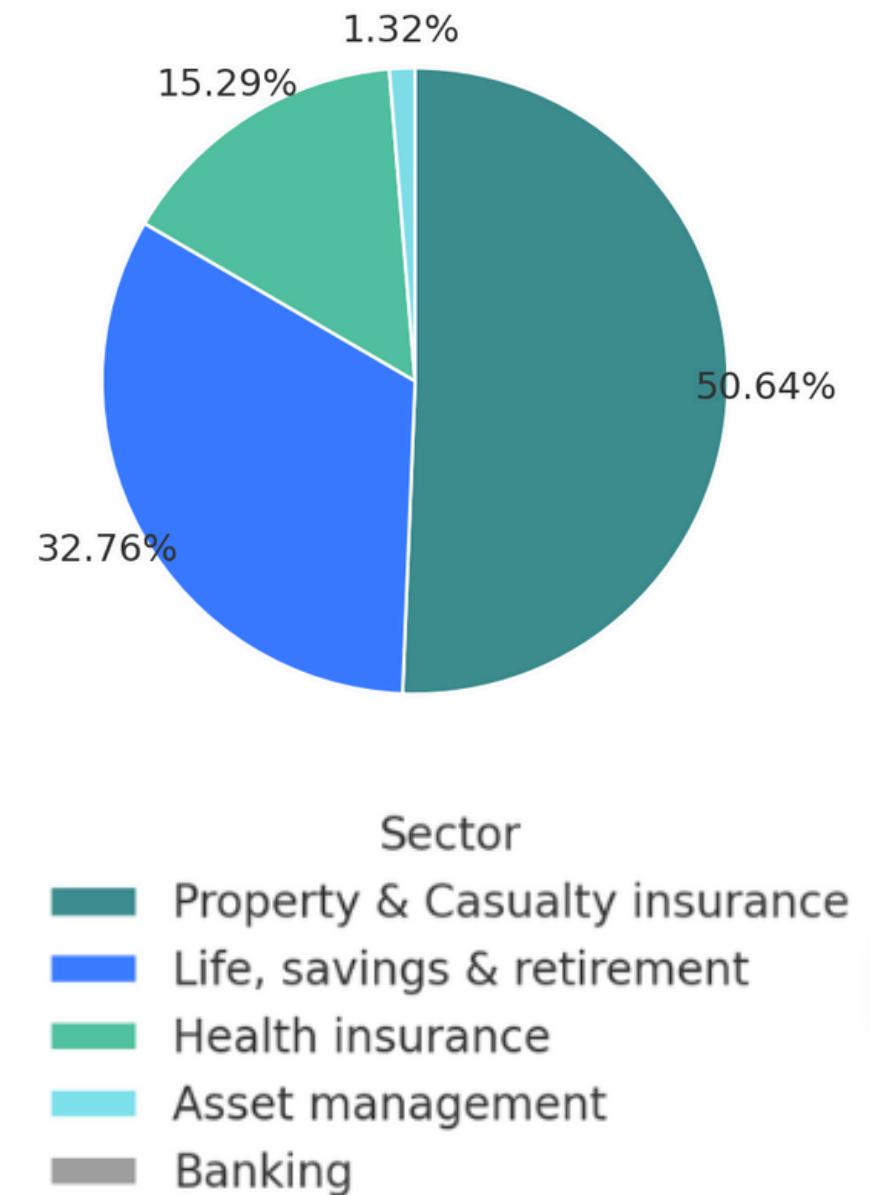
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CONTEXT

- Insurances succes =  clients  client churn
- AXA France:
 - 29% pourcent of AXA Group turnover
 - 3,000 agencies
 - 8,300 advisers
- Customer knowledge is an operational necessity
- This project takes place in the context of the Customer Management of AXA France.



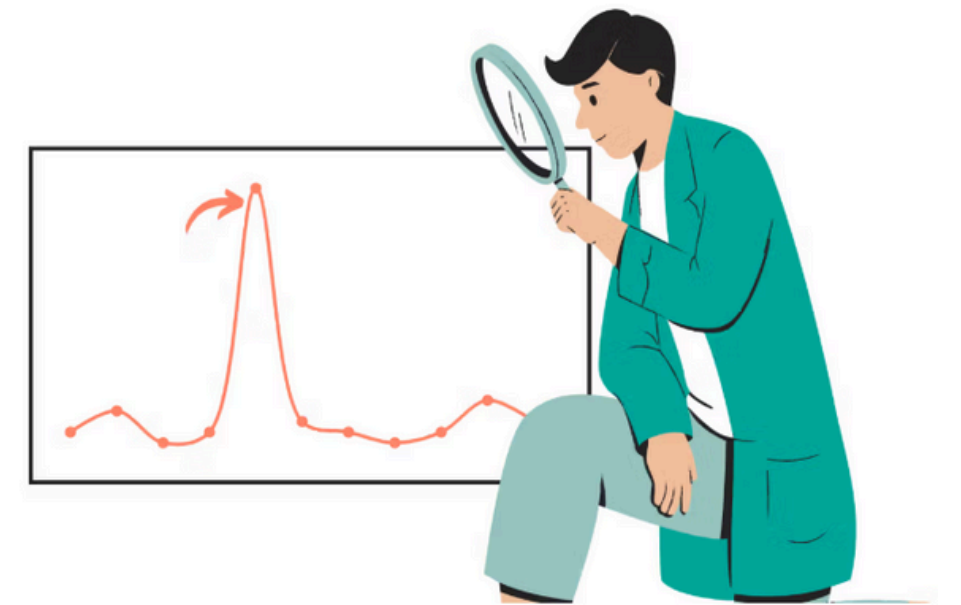
OBJECTIVE: MINIMIZATION OF CLIENT CHURN

- 2 sub-objectives:
 -  Efficient detection
 -  Interpretation / explainability of the detection
- Need of leveraging client knowledge



PROBLEM STATEMENT

- This type of tasks often lead to situations with highly imbalanced datasets:
 - Detect churn on new / recent products
 - Detect on different analysis perimeters
- Insurance standard: Generalized Linear Models (GLMs)
- Anomaly Detection: field of machine learning specialized in very imbalanced datasets.
 - Binary classification task with class-1 proportion of ~ 1% to 10%



KEY QUESTION

**TO WHAT EXTENT ANOMALY DETECTION METHODS ENABLE A
MODEL TO PERFORM BETTER ON AUC AND RECALL- CENTRIC
METRICS COMPARED TO GLMS AND CLASSICAL NEURAL
NETWORK CLASSIFIERS ?**

OVERVIEW

01 Anomaly Detection

02 Literary Review

03 Presentation of the task

04 Experiments

05 Conclusion

ANOMALY DETECTION

“An anomaly (also known as an outlier or a novelty) is typically defined as a data instance that does not conform to the expected pattern of the dataset.”,

Deep Learning for Anomaly Detection: A Review - G. Pang et al.

Definition



Different forms: Point anomalies, Conditional anomalies, Group anomalies

Properties:

- Unpredictable behavior
- Heterogeneous semantics
- Rarity => Class imbalance
- Misclassifying is costly

Key challenges



- Low detection rate
- Noise-resilient detection
- Complex anomalies detection
- Interpretability

ANOMALY DETECTION

Modeling approaches

Unsupervised:

Strong assumptions on the data distribution. Enable us to get rid of the small amount of labeled data.

01

02

03

Semi-supervised:

Normal data available but no or few labeled anomalies. Attempt to learn a strong representation of normal data in order to discriminate any sample that deviate from it.

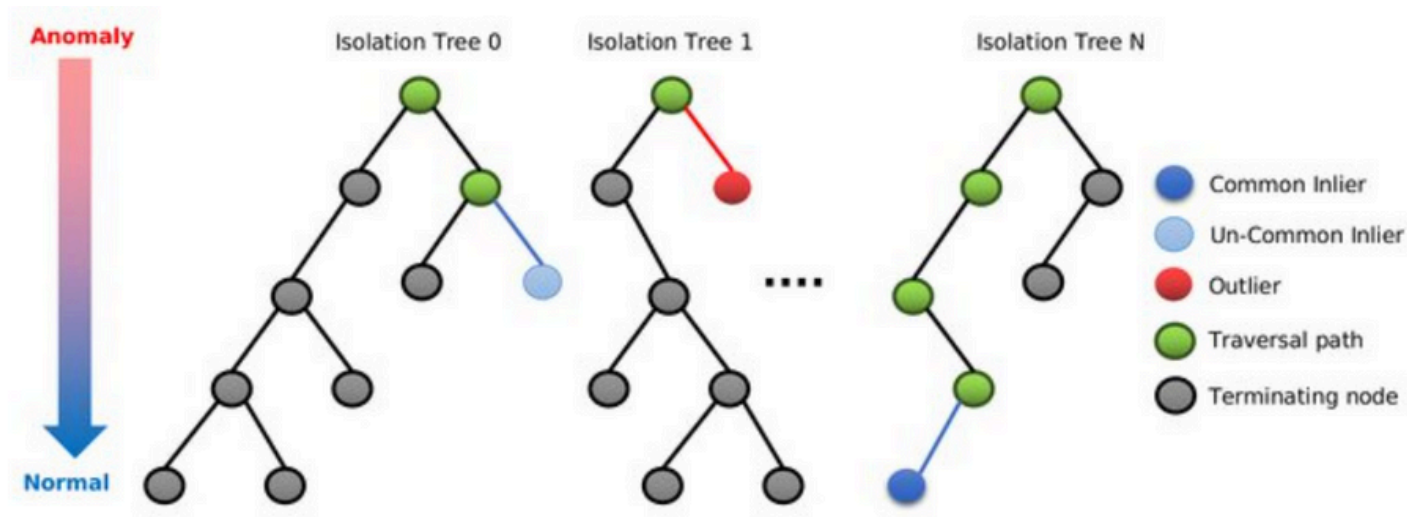
Weakly-supervised:

Few labeled anomalies available.
Combine normality learning with specific architecture / loss to take advantage of the few labeled anomalies.

LITERARY REVIEW

Anomaly Detection models

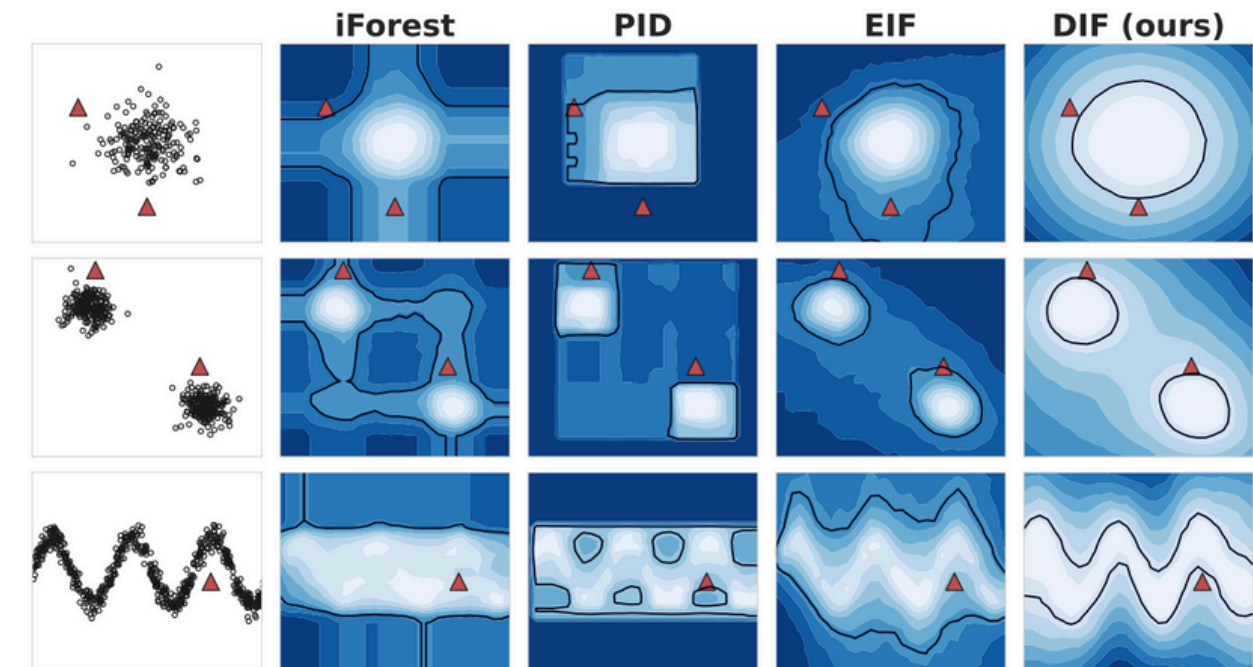
iForest



- iForest = collection of iTrees
- iTree = succession of nodes
- Node = One feature uniformly selected. One split value uniformly selected.

$$s(x) = 2^{-\frac{\mathbb{E}[h(x)]}{c(m)}}$$

Deep iForest



- r independent random mappings
- t iTrees on each projected dataset

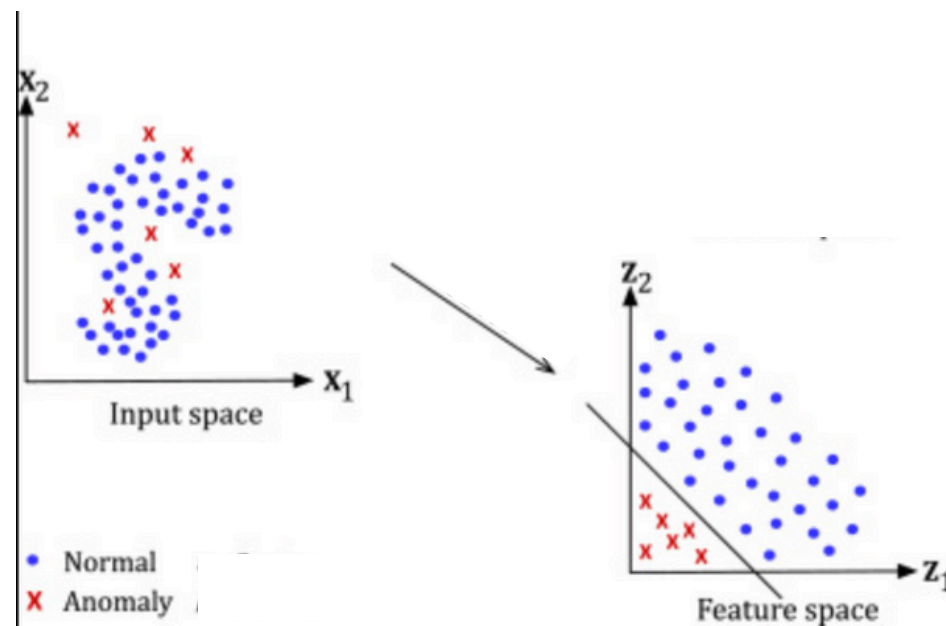
$$g(x_u | \tau_i) = \frac{1}{|p(x_u | \tau_i)|} \sum_{k \in p(x_u | \tau_i)} |x_u^{(j_k)} - \eta_k|$$

$$F_{\text{DEAS}}(\mathbf{o} | T) = 2^{-E_{\tau \in T}[|p(x_u | \tau)|]/C(T)} \times E_{\tau \in T}[g(x_u | \tau)]$$

LITERARY REVIEW

Anomaly Detection models

OC-SVM



Minimization problem:

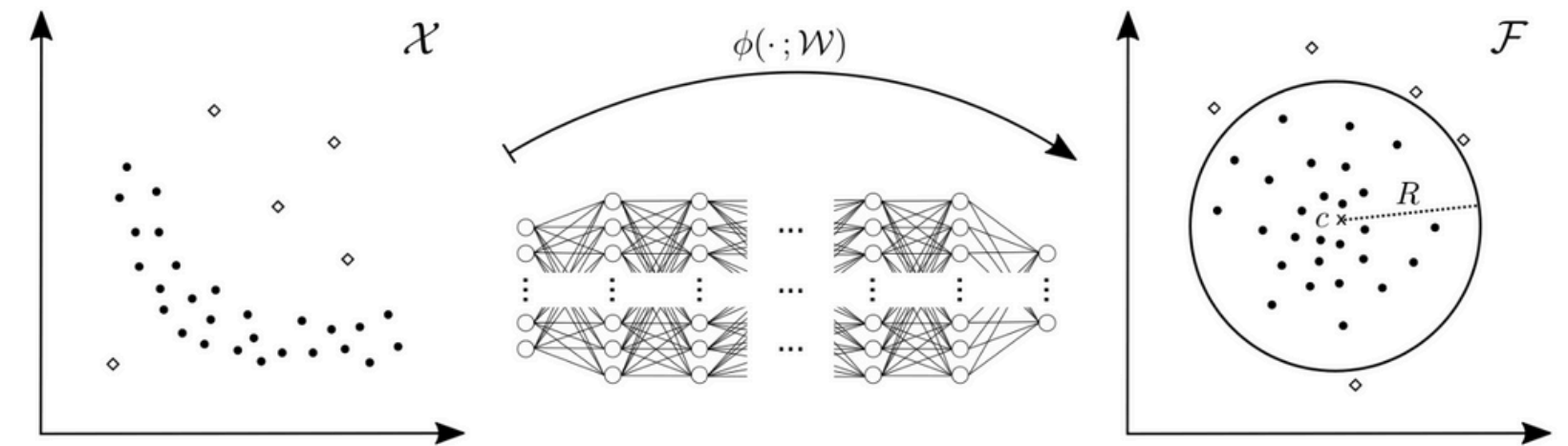
$$\min_{w, \rho, \{\xi_i\}} \frac{1}{2} \|w\|^2 + \frac{1}{\nu n} \sum_{i=1}^n \xi_i - \rho,$$

subject to $w^\top \Phi(x_i) \geq \rho - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \dots, n$

Decision function:

$$f(x) = w^\top \Phi(x) - \rho$$

Deep SAD: semi or unsupervised



- Minimization problem:

$$\min_W \frac{1}{n+m} \sum_{i=1}^n \|\phi(x_i; W) - c\|^2 + \frac{\eta}{n+m} \sum_{j=1}^m (\|\phi(\tilde{x}_j; W) - c\|^2)^{\tilde{y}_j} + \frac{\lambda}{2} \sum_{\ell=1}^L \|W^\ell\|_F^2$$

$y \in \{-1, 1\}$

- Training details:

- Pre-trained autoencoder on normal data used to initialize the neural network

- Decision function: $s(\tilde{x}) = \|\phi(x) - c\|$

LITERARY REVIEW

Anomaly Detection models

FEAWAD: weakly supervised

- Latent representation h

- Reconstruction error $r = \frac{\hat{\mathbf{x}}_i - \mathbf{x}_i}{\|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2}$

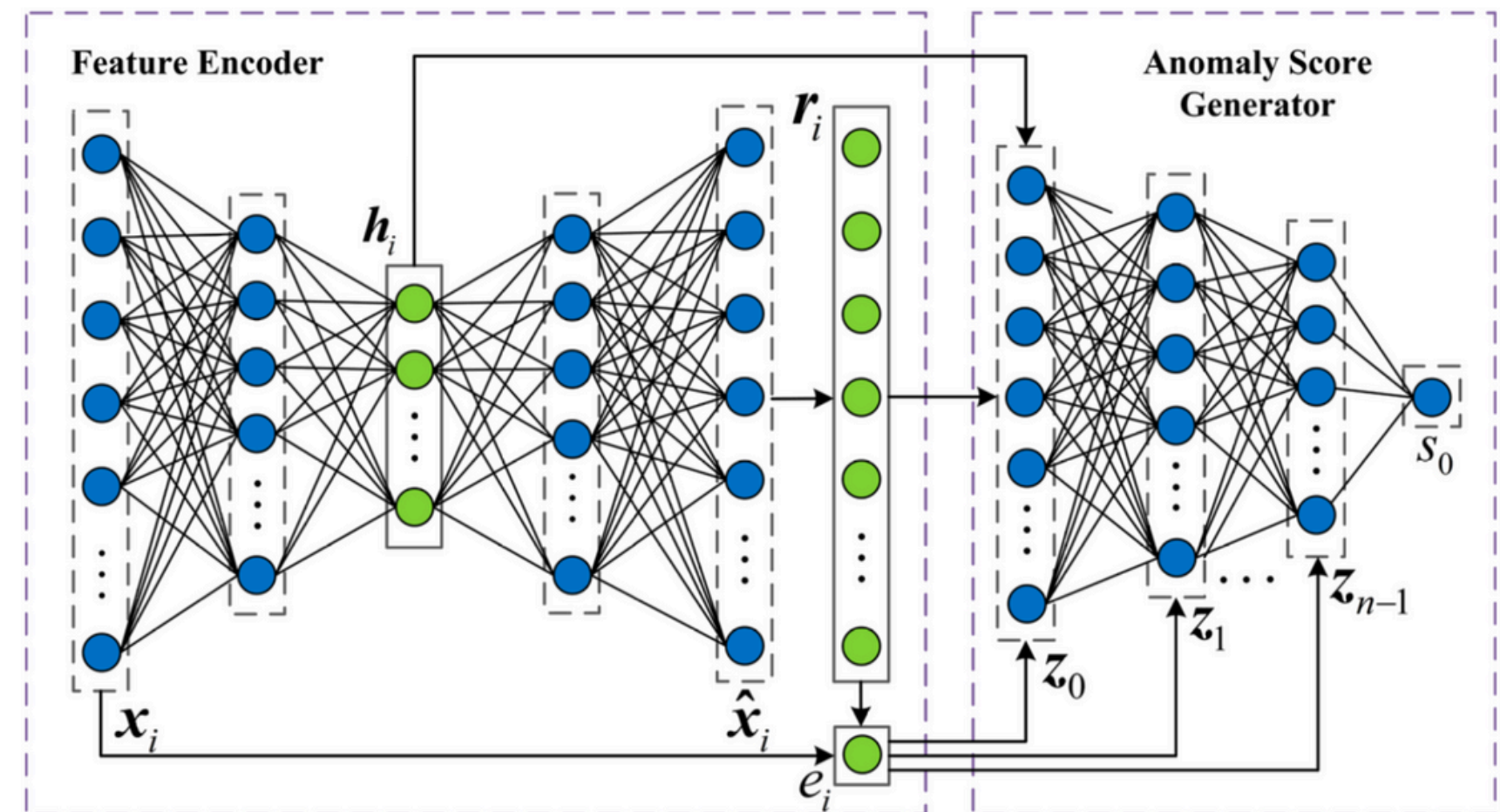
- Residual $e = \|x - \hat{x}\|^2$

- Final objective:

$$\mathcal{L}(\Theta_e, \Theta_g) = \mathcal{L}_d(\Theta_e, \Theta_g) + \lambda \mathcal{L}_e(\Theta_e)$$

$$\mathcal{L}_d = \sum_i (1 - y_i) |\phi(x_i)| + y_i \max(0, a_0 - \phi(x_i))$$

$$\mathcal{L}_e = \sum_i (1 - y_i) e_i + y_i \max(0, a_0 - e_i)$$



LITERARY REVIEW

Anomaly Detection models

- Surrogate dataset:

$$\mathcal{P} = \left\{ \left(\mathbf{x}_i, \mathbf{x}_j, y_{\{\mathbf{x}_i, \mathbf{x}_j\}} \right) \mid \mathbf{x}_i, \mathbf{x}_j \in \mathcal{X} \text{ and } y_{\{\mathbf{x}_i, \mathbf{x}_j\}} \in \mathbb{N} \right\}$$

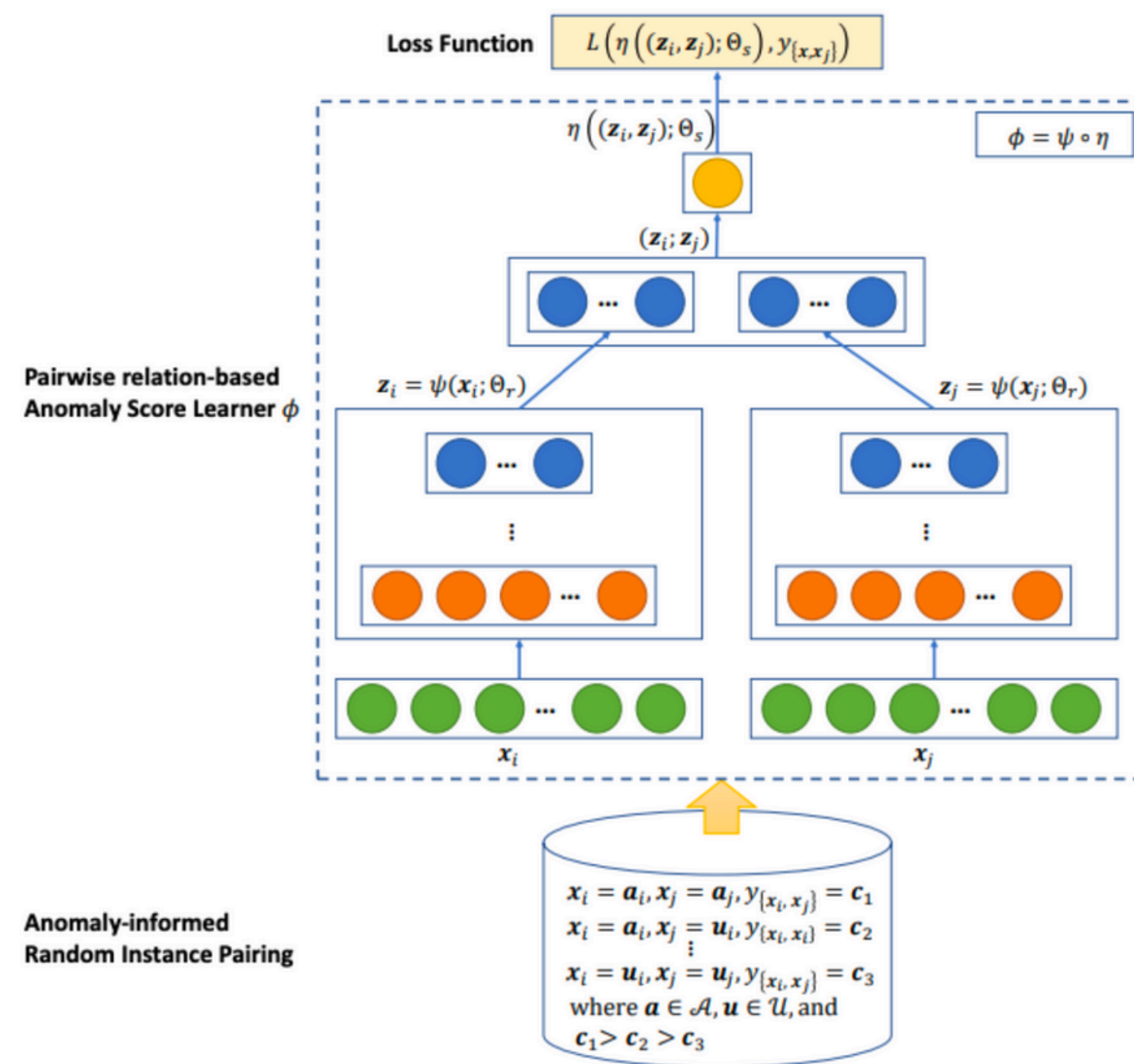
- Surrogate labels:

$$y_{\{a,a\}} = 8, y_{\{a,u\}} = 4 \text{ and } y_{\{u,u\}} = 0$$

- We get $(N + K)^2$ pairs:
 - K^2 $\{a, a\}$ pairs
 - $2KN$ $\{u, a\}$ pairs (i.e. $\{a, u\}$)
 - N^2 $\{u, u\}$ pairs

- Minimization problem:

$$\arg \min_{\Theta} \frac{1}{|\mathcal{B}|} \sum_{(\mathbf{x}_i, \mathbf{x}_j, y_{\{\mathbf{x}_i, \mathbf{x}_j\}}) \in \mathcal{B}} \left| y_{\{\mathbf{x}_i, \mathbf{x}_j\}} - \phi((\mathbf{x}_i, \mathbf{x}_j); \Theta) \right| + \lambda R(\Theta)$$



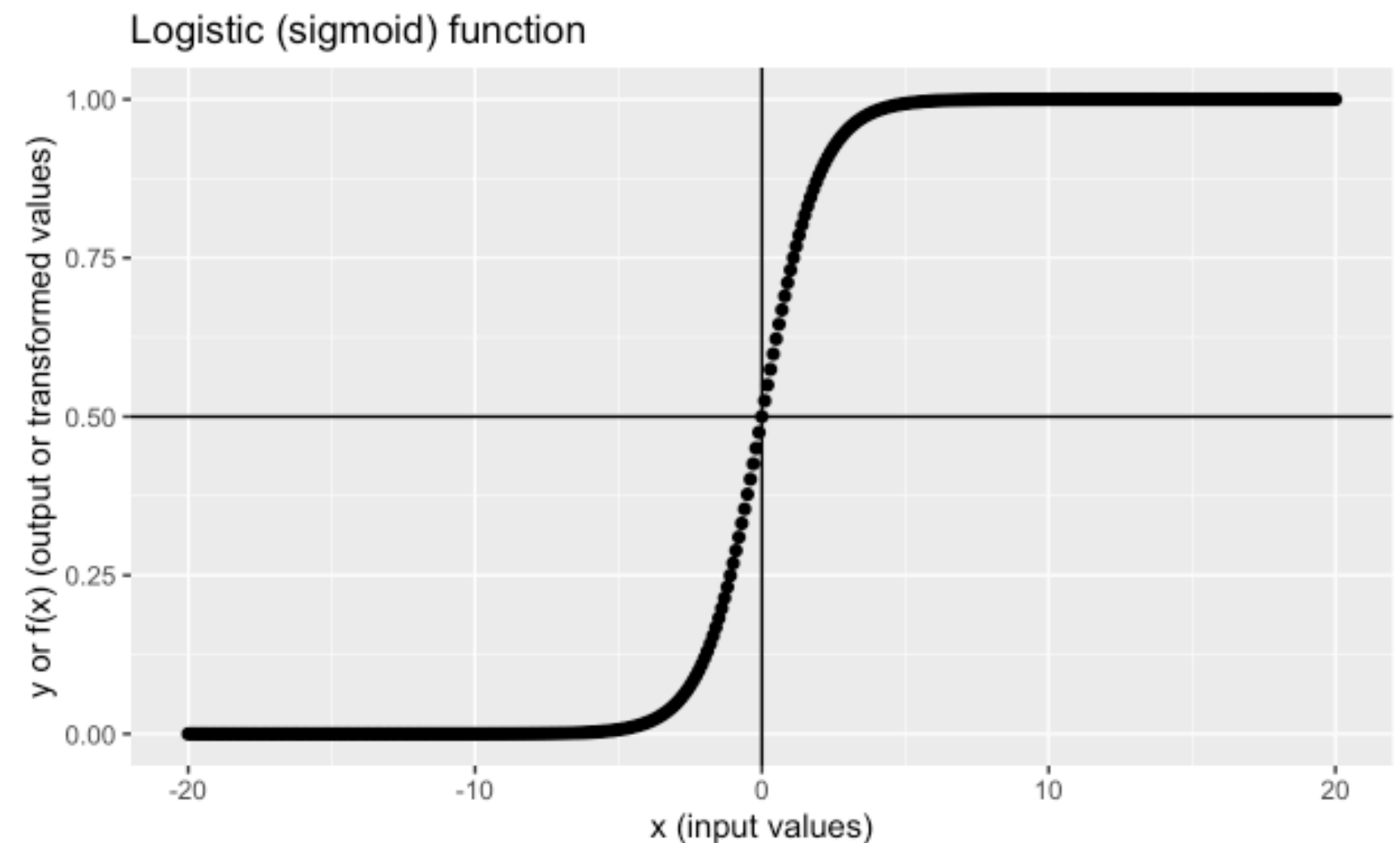
LITERARY REVIEW

GLMs: Logistic Regression example

- Link function $f(x) = \frac{1}{1 + e^{-x}}$
- Linear predictor $y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} = \mathbf{x}_i^T \boldsymbol{\beta}$.
- The final predictor:

$$\Pr(Z_i = 1) = \frac{1}{1 + \exp(-\mathbf{x}_i^T \boldsymbol{\beta})}$$

- With Z following a binomial law



PRESENTATION OF THE TASK

Objectives

- Efficient detection within 3 months
 - a. Maximize TP and TN.
 - b. We prefer to minimize FN rather than FP.
- Minimize FP $\iff$ Maximize Recall
 - Recall: $\frac{TN}{TN+FN}$
- Minimize FP $\iff$ Maximize Precision
 - Precision: $\frac{TP}{TP+FP}$

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

PRESENTATION OF THE TASK

The dataset

- Perimeter:
 - Auto insurance contracts subscribed between January 2023 and January 2025 (to avoid concept drift).
- One data point = a snapshot of a contract at a given date.
- For a given contract: one snapshot for every 3 months from the subscription to the termination (or the max date)
- Target = 1 if the contract is terminated within the 3 months following the snapshot date, else 0.

ID	N_Contrat	Snapshot	Client_Age	Contribution	Target
1	1	03-2023	481	347	0
2	1	06-2023	484	347	0
3	1	09-2023	487	361	1
4	2	09-2025	376	80	0
5	2	12-2025	379	80	0
6	3	01-2024	603	660	0
...	...	...	...	...	...

- $n = 6,746,342$ samples.
- $d = 53$ features
- % class 1 = 2,3%

03

PRESENTATION OF THE TASK

The dataset

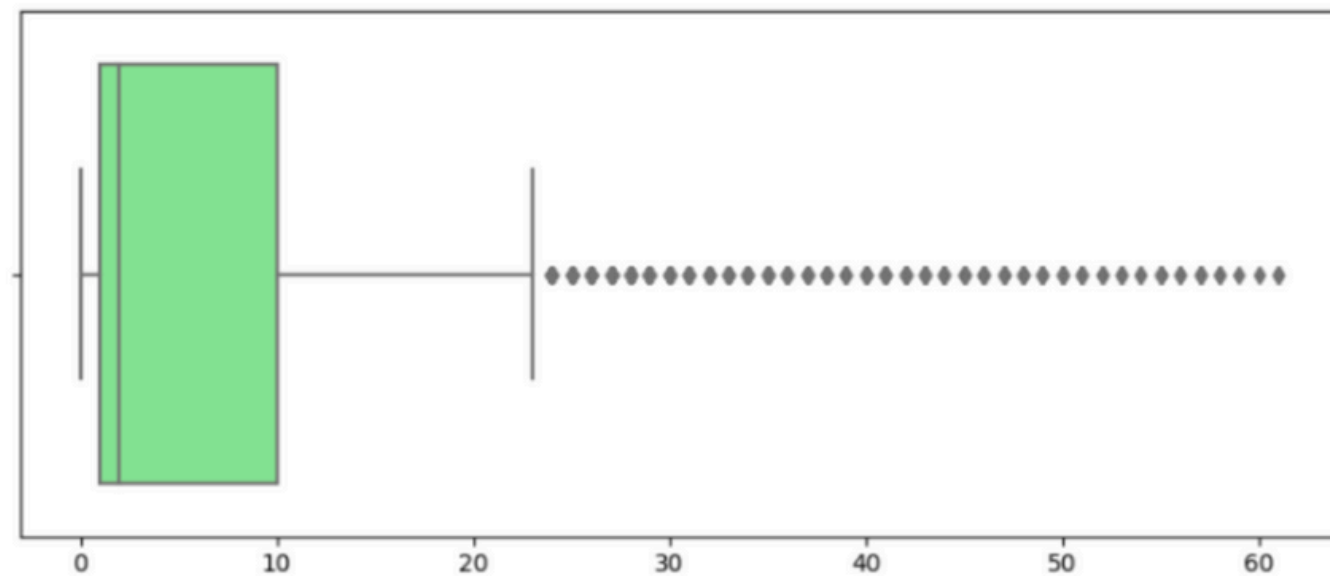


Figure : Box plot of **Client's Seniority in months** feature

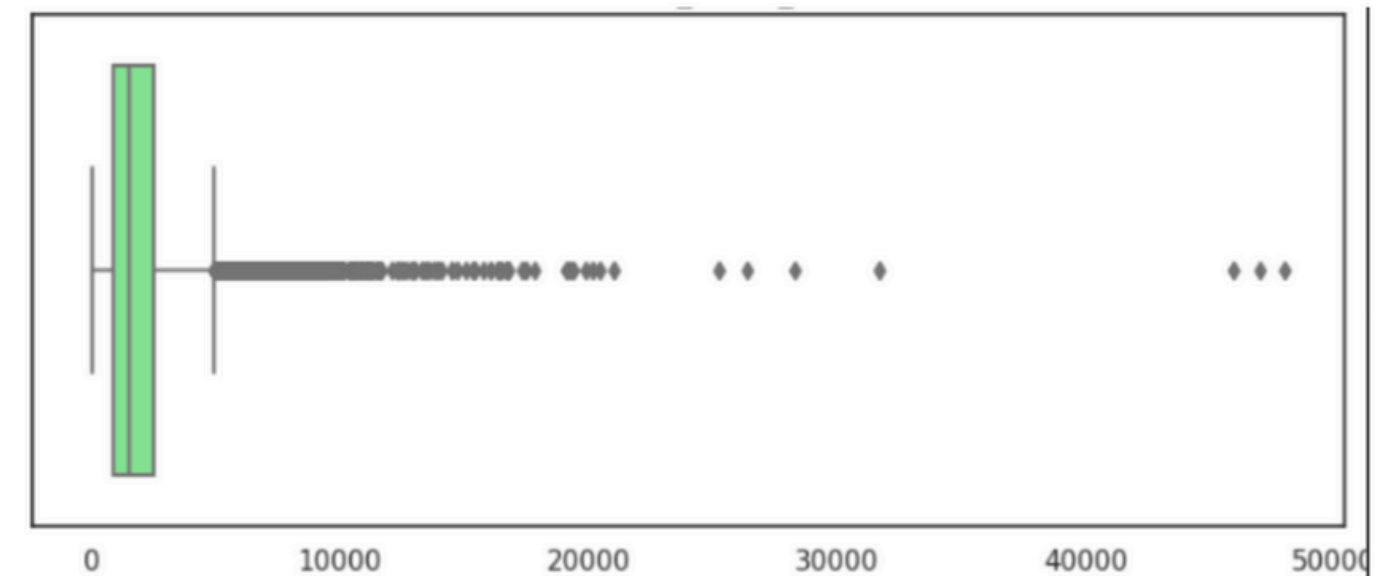


Figure : Box plot of **Client's Total Contribution** feature

PRESENTATION OF THE TASK

The dataset

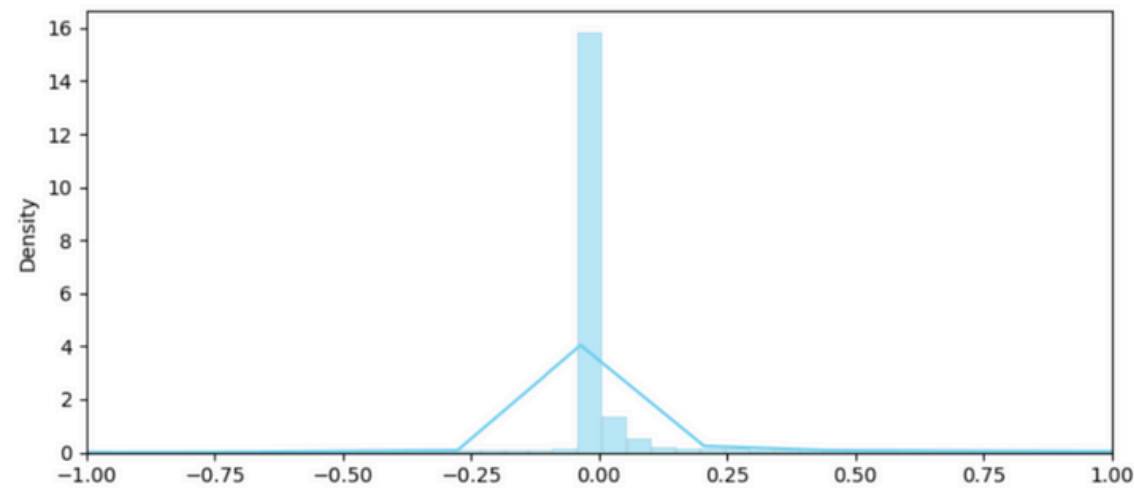


Figure 14: Distribution of **Evolution of Client's Auto Contribution from $t-3$ to t** in **class 0**

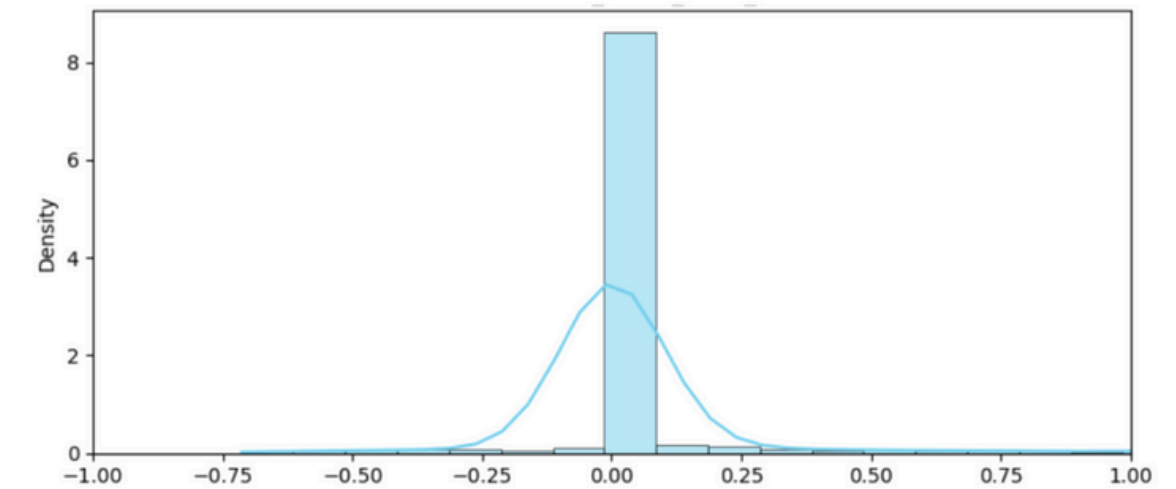


Figure 15: Distribution **Evolution of Client's Auto Contribution from $t-3$ to t** in **class 1**

PRESENTATION OF THE TASK

The dataset

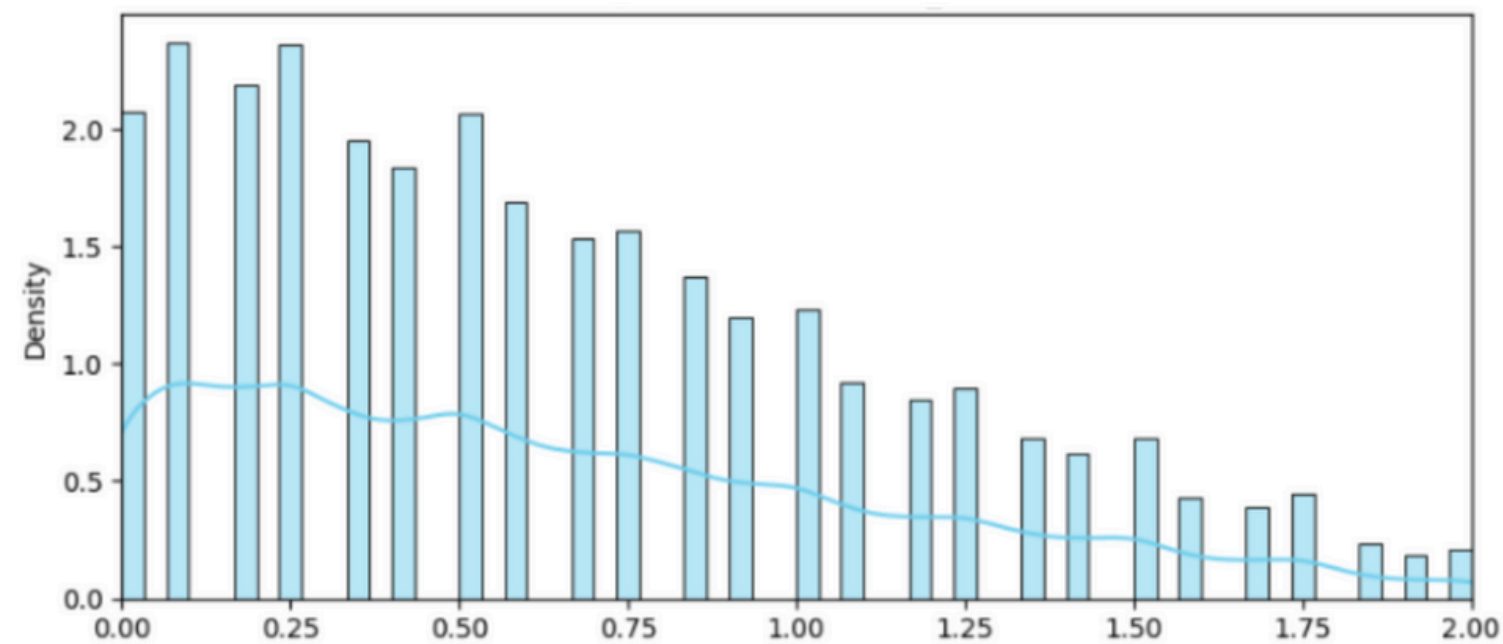


Figure : Distribution of **Contract's seniority** in **class 0**

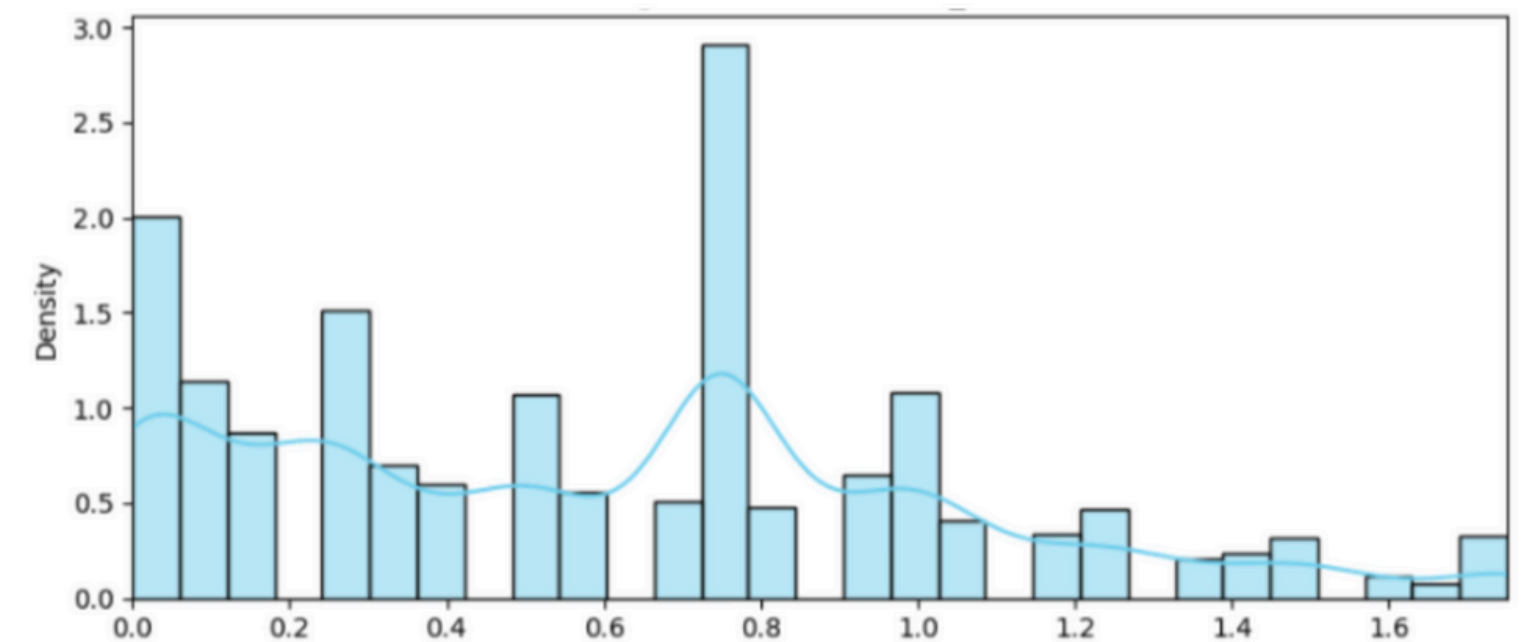


Figure : Distribution **Contract's Seniority** in **class 1**

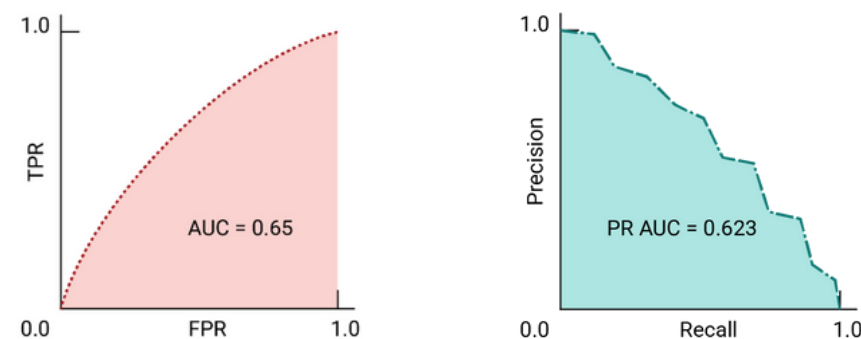
PRESENTATION OF THE TASK

Experiments settings



Evaluating

- AUC-ROC, AUC-PR



- f1-score
- Recall
- Precision

PRESENTATION OF THE TASK

Experiments settings



Implementations

- 200k stratified subsets
- 5 runs
- We report mean +- standard deviation
- We get the recall and precision associated to the best f1-score found on the test data after training.
- Models are either self implemented, from the original github of the papers or from the deepod library

EXPERIMENTS

Anomaly Detection models

How do we need to proceed ?

Phase 02

Empirically understand how the architecture of the selected models work and can be leveraged.

Phase 01

Understand the types of AD models that best fit our task.

Phase 03

Compare the strengths and weaknesses of the selected models with the comparison models.

EXPERIMENTS

Phase 01

Understand the types of AD models that best fit our task.

- Weakly > Semi > Unsupervised
- Weakly models:
 - FEAOWAD is better than DevNet in AUC-PR and f1-score, like in the literature.
 - PReNet: big gap with other models.
Demonstrates a more balanced f1-score but still favors recall. Better precision = less false positives = generalizes better to unseen data.
- Confirm improvements of the literature.

Models	Type	Global Performance		Operational Performance		
		AUC-ROC	AUC-PR	Best f1-score	Recall	Precision
DevNet	Weakly	0.629±0.014	0.041±0.003	0.053±0.00	0.85±0.00	0.028±0.00
FEAWAD	Weakly	0.607±0.016	0.045±0.005	0.060±0.00	0.641±0.03	0.03±0.00
Deep SVDD	Unsupervised	0.534±0.004	0.028±0.001	0.051±0.01	0.68±0.03	0.03±0.01
Deep SAD	Semi	0.561±0.014	0.038±0.007	0.07±0.01	0.16±0.04	0.04±0.01
Deep iForest	Unsupervised	0.556±0.032	0.031±0.004	0.07±0.01	0.46±0.05	0.05±0.04
PReNet	Weakly	0.651±0.005	0.062±0.007	0.109±0.01	0.151±0.04	0.109±0.05

Table : Performances of deep anomaly detection models (mean ± standard deviation) on 5 runs. Global performance metrics enable to measure how well the models behave over all thresholds. Operational Performance enable to measure the best f1-score, and its associated recall and precision.

EXPERIMENTS

Phase 02

Empirically understand how the architecture of the selected models work and can be leveraged.

Ablative implementation:

- MLP: pre-trained MLP, binary crossentropy
- AD_MLP: pre-trained MLP with only normal data, binary crossentropy and h , e , r features from FEAAD.

- FEAAD

Other models:

- PReNet
- Logistic Regression

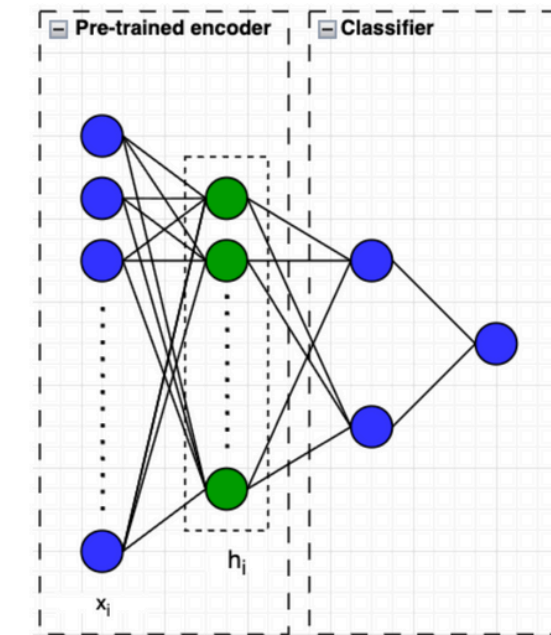


Figure : Architecture of the basic MLP model with a pre-trained encoder. The number of layers is for illustration purposes.

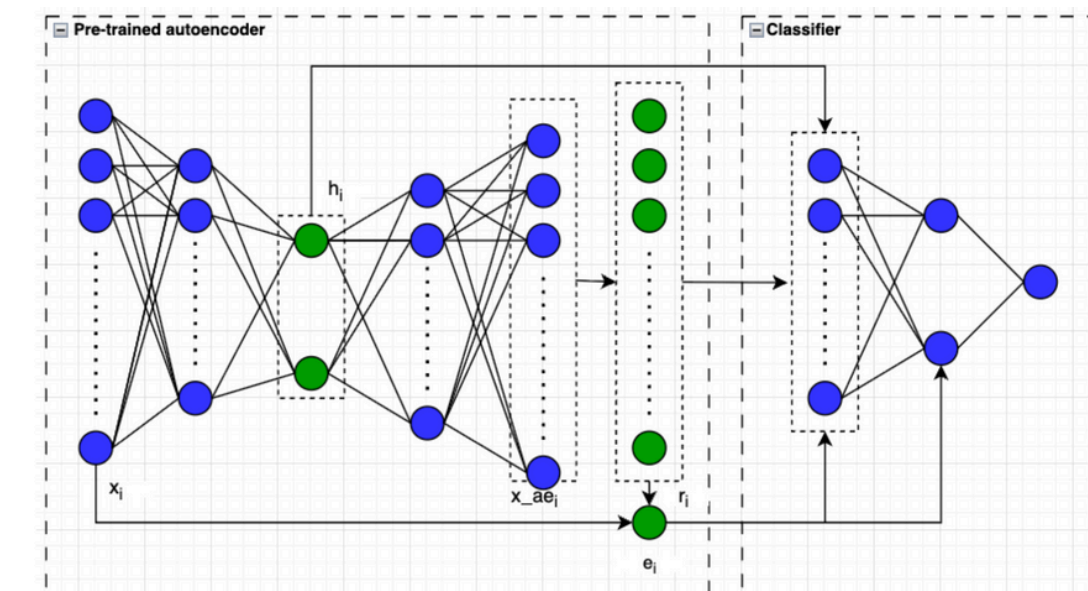


Figure : Architecture of AD_MLP with representations features h , e , r from FEAAD model. The number of layers is for illustration purposes.

EXPERIMENTS

Phase 02

Empirically understand how the architecture of the selected models work and can be leveraged.

- Each of FEAAD features increase Recall / decrease Precision.
- AD_MLP demonstrate better performances
 - On our task: e, r increase AUC-ROC by 2% and AUC-PR by 1,5%
 - In FEAAD paper: e, r increase AUC-ROC by 2,7% and AUC-PR by 6,8%.
- Max-margin-like loss and end-to-end batch strategy deteriorate performances
 - f1-score not balanced: Recall too high and Precision too low
- Basic MLP is better than Log. Reg.

Models	Global Performance		Operational Performance		
	AUC-ROC	AUC-PR	Best f1-score	Recall	Precision
MLP	0.643±0.008	0.073±0.009	0.114±0.02	0.105±0.00	0.247±0.11
AD_MLP	0.656±0.006	0.078±0.011	0.120±0.02	0.125±0.01	0.196±0.10
FEAD	0.607±0.016	0.045±0.005	0.060±0.00	0.641±0.03	0.03±0.00
PReNet	0.651±0.005	0.062±0.007	0.109±0.01	0.151±0.04	0.109±0.05
Log. Reg.	0.634±0.009	0.069±0.009	0.112±0.01	0.096±0.04	0.267±0.10

Table : Performances of each model (mean ± standard deviation) on 5 runs. Global performance metrics enable to measure how well the models behave over all thresholds. Operational Performance enable to measure the best best f1-score, and its associated recall and precision.

EXPERIMENTS

Phase 02

Empirically understand how the architecture of the selected models work and can be leveraged.

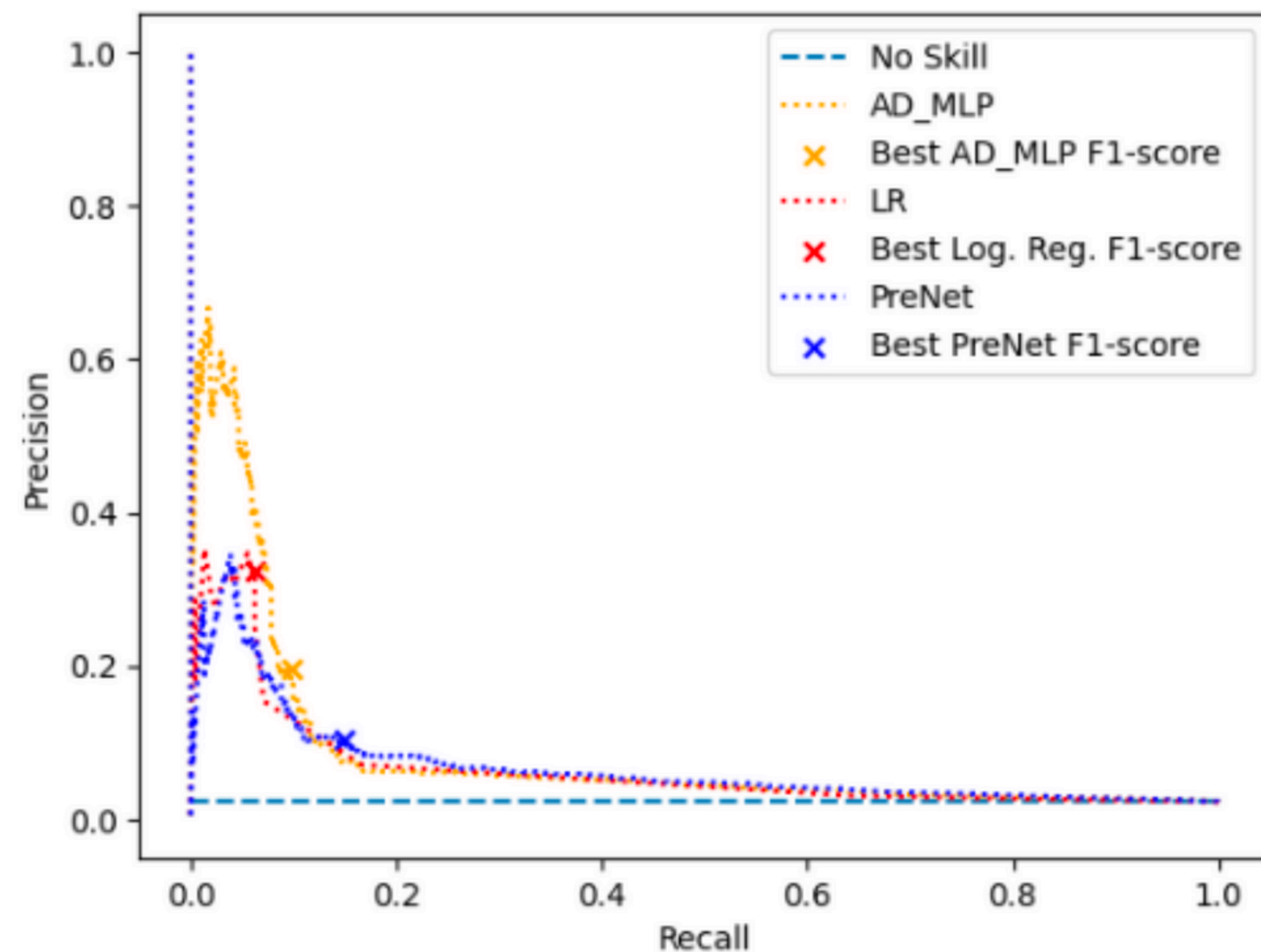


Figure 20: Example of precision curves of the models on a test dataset of $\approx 20,000$ samples.

EXPERIMENTS

Phase 03

Compare the strengths and weaknesses of the selected models with the comparison models.

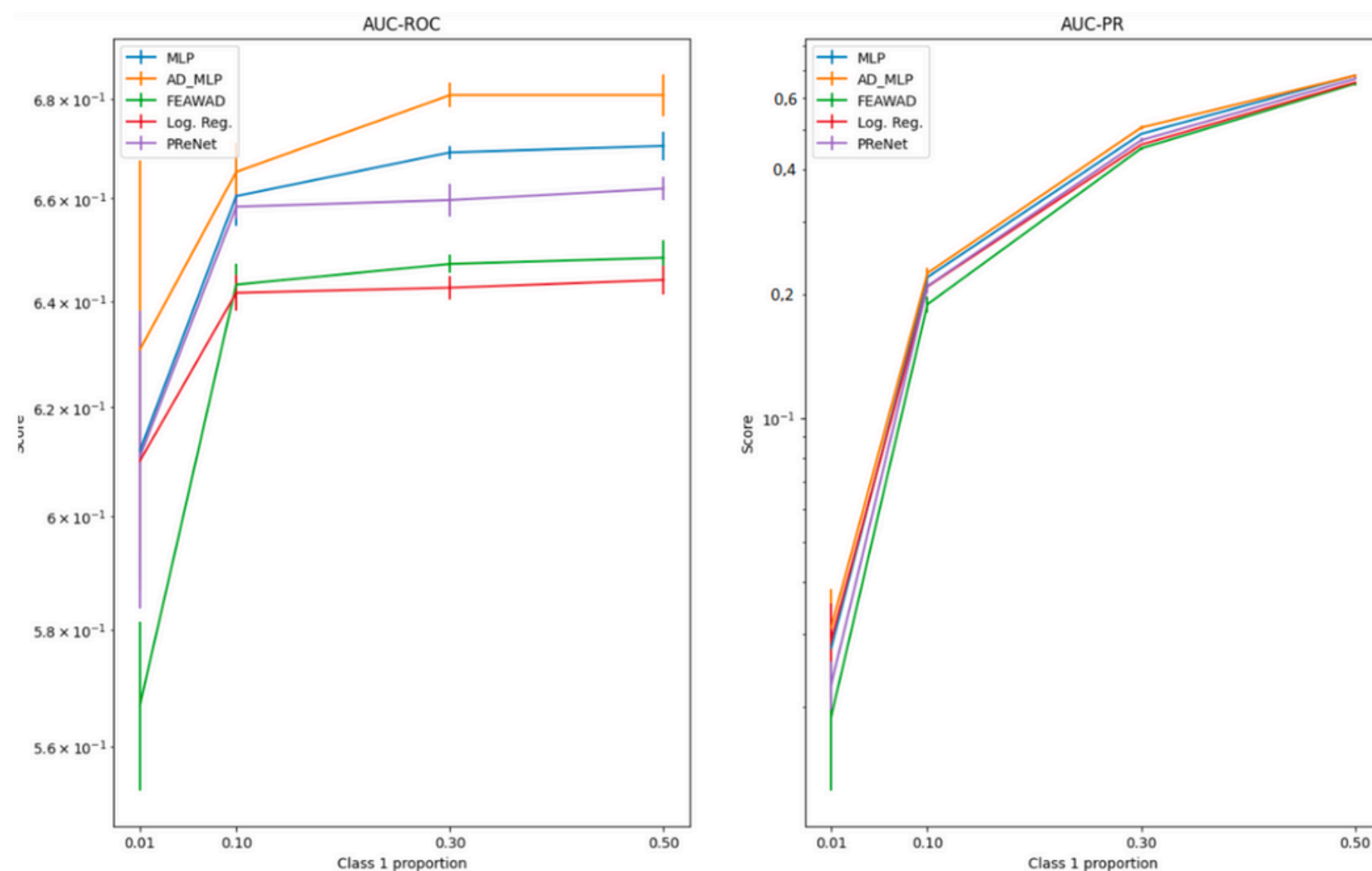


Figure 21: Global performances of the models with variation in the class proportions. We use a log scale for Y axis.

EXPERIMENTS

Phase 03

Compare the strengths and weaknesses of the selected models with the comparison models.

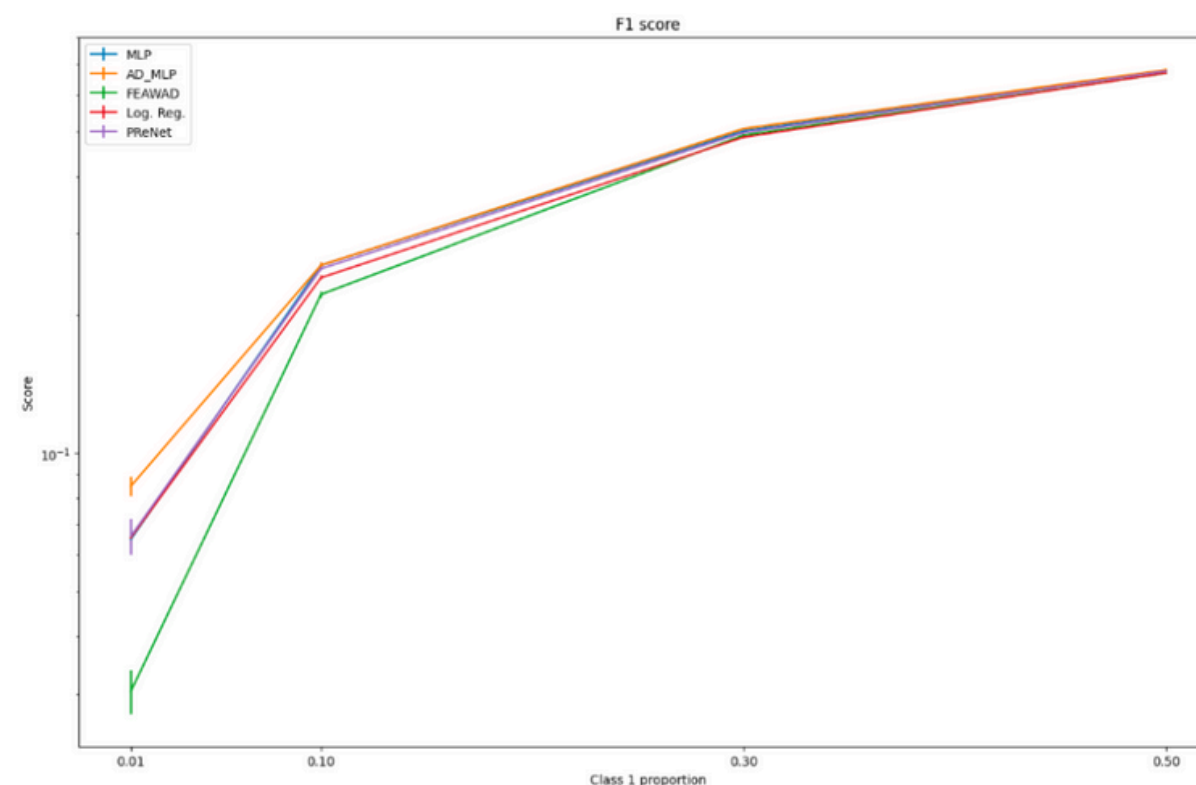


Figure 22: F1 score of the models with variation in the class proportions. We use a log scale for Y axis.

EXPERIMENTS

Phase 03

Compare the strengths and weaknesses of the selected models with the comparison models.

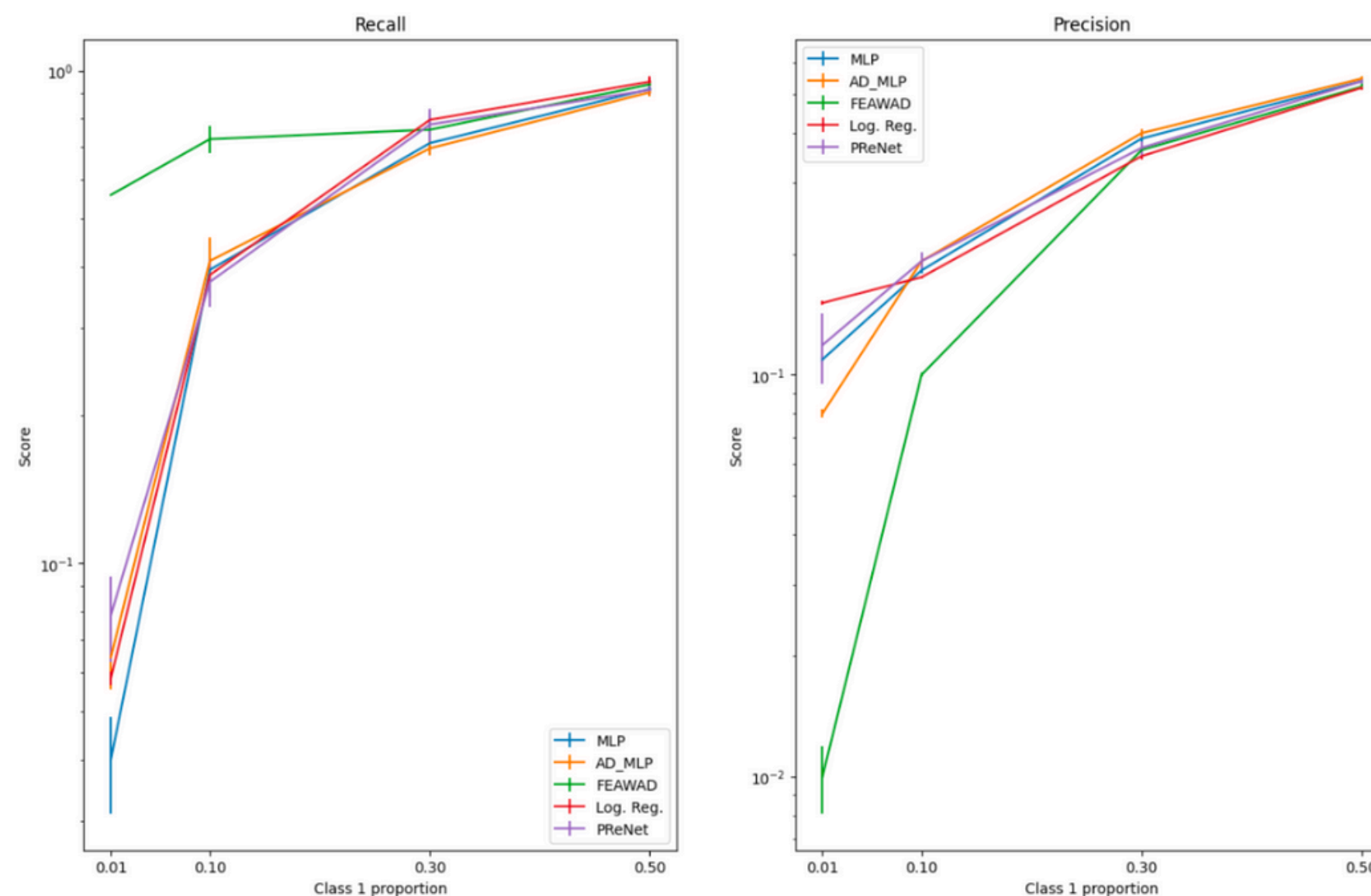


Figure 23: Operational performances-Recall and Precision-of the models with variation in the class proportions. We use a log scale for Y axis.

CONCLUSION

- 01 Carried out work: weakly supervised > semi supervised > unsupervised. FEAWAD features favor recall for low levels of class proportion. We managed to achieve the best performances for all class proportion levels with AD_MLP which leverage some FEAWAD features. Anomaly detection features exploit their full potential with highly imbalanced datasets.
- 02 Limitations: Anomaly detection models assume that the data is perfectly clean. We have bias in the data. Furthermore, we work with a recent data lab and there are still data quality issues.
- 03 Possible future works: Adapt architectures for mixed data (categorical and numerical). Consider the anomalies as group anomalies (One contract is a group of snapshots that together represent a terminated contract even if each snapshot is not necessarily a terminated contract). New methods with attention mechanism are taking place.

THANK YOU FOR LISTENING

